

**johnston-linear-matrix-algebra: residuals**

equations measured 2026-09-05T12:23:12Z against the model of 2026-09-01

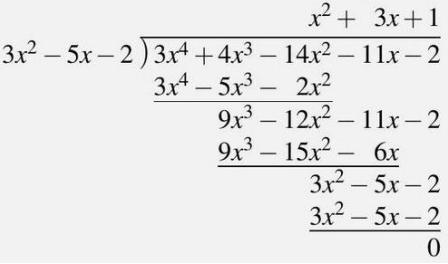
## Corrected (8)

[illegible]

[illegible]

| Identifier                                                                                                                                                                                                                                                                                                                       | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Rendered                                                                                                                                                                                                                                                                                                                                                                           | Scan image                                                                                                                                                                                                                                                                                                                                                                                               |
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| johnston-linear-matrix-augmented<br>EQ0528 (was)                                                                                                                                                                                                                                                                                 | 212  | ■0.001 | $\begin{array}{r} \leftarrow \begin{array}{c} z \\ R_1 \end{array} \rightarrow \begin{array}{c} S_2 \quad S_3 \\ R_3 + R_4 \\ 0 \\ 0 \\ 0 \end{array} \left  \begin{array}{cccccc c} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{array} \right. \left[ \begin{array}{c ccc ccc} 1 & 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 1 & -2 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 1 \end{array} \right] \begin{array}{c} 2 \\ \text{pivot} \end{array} \end{array}$                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                          |
| johnston-linear-matrix-augmented<br>EQ0528 (now)                                                                                                                                                                                                                                                                                 | 212  | —      | $\begin{array}{l} \rightarrow \begin{array}{c} z \\ R_1 \end{array} \rightarrow \begin{array}{c} S_2 \quad S_3 \\ R_3 + R_4 \\ R_4 + R_4 \end{array} \left[ \begin{array}{c ccc ccc} 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 & -1 & -1 & -1 & -1 \\ 0 & 0 & 0 & 1 & -2 & 1 & -1 & -2 \end{array} \right] \curvearrowright \text{pivot here} \xrightarrow{\begin{array}{c} -R_4 \\ R_1 - R_4 \\ R_3 + R_4 \end{array}} \left[ \begin{array}{c ccc ccc} 1 & 0 & 0 & 1 & -2 & 1 & 0 & -2 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & -1 & 2 & -1 & 1 & 2 \end{array} \right] \end{array}$ |                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                          |
| basis: inferred / ink                                                                                                                                                                                                                                                                                                            |      |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                          |
| johnston-linear-matrix-augmented<br>EQ0543 (was)                                                                                                                                                                                                                                                                                 | 218  | ■0.103 | $\begin{array}{l} \operatorname{operatorname{minimize}}: \quad & 3y_1 + 3y_2 + 3y_3 + 3y_4 + 3y_5 \\ \text{subject to: } & y_1 + y_4 + y_5 \geq 1 \\ & y_1 + y_2 + y_5 \geq 1 \\ & y_1 + y_2 + y_3 \geq 1 \\ & y_2 + y_3 + y_4 \geq 1 \\ & y_3 + y_4 + y_5 \geq 1 \\ & y_1, \quad y_2, \quad y_3, \quad y_4, \quad y_5 \geq 0 \end{array}$                                                                                                                                                                                                                                                                                                                            | $\begin{array}{l} \operatorname{minimize} : \quad 3y_1 + 3y_2 + 3y_3 + 3y_4 + 3y_5 \\ \text{subject to: } \quad y_1 + y_4 + y_5 \geq 1 \\ \quad \quad \quad y_1 + y_2 + y_5 \geq 1 \\ \quad \quad \quad y_1 + y_2 + y_3 \geq 1 \\ \quad \quad \quad y_2 + y_3 + y_4 \geq 1 \\ \quad \quad \quad y_3 + y_4 + y_5 \geq 1 \\ y_2, \quad y_3, \quad y_4, \quad y_5 \geq 0 \end{array}$ | <div>minimize: <math>3y_1 + 3y_2 + 3y_3 + 3y_4 + 3y_5</math></div> <div>subject to: <math>y_1 + y_4 + y_5 \geq 1</math></div> <div><math>y_1 + y_2 + y_5 \geq 1</math></div> <div><math>y_1 + y_2 + y_3 \geq 1</math></div> <div><math>y_2 + y_3 + y_4 \geq 1</math></div> <div><math>y_3 + y_4 + y_5 \geq 1</math></div> <div><math>y_1, \quad y_2, \quad y_3, \quad y_4, \quad y_5 \geq 0</math></div> |
| <div>Conf. MathPix confidence: <span>■</span> ≥0.9 <span>■</span> 0.5-0.9 <span>■</span> &lt;0.5 <span>—</span> no value</div> <div>Residual render vs scan (inkdrill): <span>●</span> C component <span>●</span> W weak <span>●</span> S stable <span>●</span> N noise <span>●</span> K clean <span>●</span> not measured</div> |      |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                          |

| Identifier                                                                                                                                                                                                                                                                                                          | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| johnston-linear-matrix-algebra-EQ0543 (now)                                                                                                                                                                                                                                                                         | 218  | —      | $v = (6, 3, 3), (-2, 0, 6) = (8, 3, 0)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $v = (6, 3, 3), (-2, 0, 6) = (8, 3, 0)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <div>minimize: <math>3y_1 + 3y_2 + 3y_3 + 3y_4 + 3y_5</math><br/>subject to: <math>y_1 + y_4 + y_5 \geq 1</math><br/><math>y_1 + y_2 + y_5 \geq 1</math><br/><math>y_1 + y_2 + y_3 \geq 1</math><br/><math>y_2 + y_3 + y_4 \geq 1</math><br/><math>y_3 + y_4 + y_5 \geq 1</math><br/><math>y_1, y_2, y_3, y_4, y_5 \geq 0</math></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| basis: inferred / ink                                                                                                                                                                                                                                                                                               |      |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| johnston-linear-matrix-algebra-EQ0748 (was)                                                                                                                                                                                                                                                                         | 283  | ■0.116 | $\begin{aligned} & \left[ \begin{array}{lll} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{array} \right] \xrightarrow[\begin{array}{l} R_3 - R_1 \end{array}]{\begin{array}{l} R_2 - R_1 \end{array}} \left[ \begin{array}{lll} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 2 & 8 \end{array} \right] \\ & \xrightarrow[\begin{array}{l} R_3 - 2R_2 \end{array}]{\begin{array}{l} R_1 - R_2 \\ R_3 - 2R_2 \end{array}} \left[ \begin{array}{lll} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{array} \right] \\ & \xrightarrow[\begin{array}{l} R_2 - 3R_3 \end{array}]{\frac{1}{2}R_3} \left[ \begin{array}{lll} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{array} \right] \xrightarrow[\begin{array}{l} R_1 + 2R_3 \\ R_2 - 3R_3 \end{array}]{\begin{array}{l} R_1 + 2R_3 \\ R_2 - 3R_3 \end{array}} \left[ \begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]. \end{aligned}$ | $\begin{aligned} & \left[ \begin{array}{lll} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{array} \right] \xrightarrow[\begin{array}{l} R_3 - R_1 \end{array}]{\begin{array}{l} R_2 - R_1 \end{array}} \left[ \begin{array}{lll} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 2 & 8 \end{array} \right] \xrightarrow[\begin{array}{l} R_3 - 2R_2 \end{array}]{\begin{array}{l} R_1 - R_2 \\ R_3 - 2R_2 \end{array}} \left[ \begin{array}{lll} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{array} \right] \\ & \xrightarrow[\begin{array}{l} R_2 - 3R_3 \end{array}]{\frac{1}{2}R_3} \left[ \begin{array}{lll} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{array} \right] \xrightarrow[\begin{array}{l} R_1 + 2R_3 \\ R_2 - 3R_3 \end{array}]{\begin{array}{l} R_1 + 2R_3 \\ R_2 - 3R_3 \end{array}} \left[ \begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]. \end{aligned}$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| johnston-linear-matrix-algebra-EQ0748 (now)                                                                                                                                                                                                                                                                         | 283  | —      | $\mathbf{w} = (6, 3, -3) - (-2, 0, 6) = (8, 3, -9).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\mathbf{w} = (6, 3, -3) - (-2, 0, 6) = (8, 3, -9).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\begin{aligned} & \left[ \begin{array}{lll} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{array} \right] \xrightarrow[\begin{array}{l} R_3 - R_1 \end{array}]{\begin{array}{l} R_2 - R_1 \end{array}} \left[ \begin{array}{lll} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 2 & 8 \end{array} \right] \xrightarrow[\begin{array}{l} R_3 - 2R_2 \end{array}]{\begin{array}{l} R_1 - R_2 \\ R_3 - 2R_2 \end{array}} \left[ \begin{array}{lll} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{array} \right] \\ & \xrightarrow[\begin{array}{l} R_2 - 3R_3 \end{array}]{\frac{1}{2}R_3} \left[ \begin{array}{lll} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{array} \right] \xrightarrow[\begin{array}{l} R_1 + 2R_3 \\ R_2 - 3R_3 \end{array}]{\begin{array}{l} R_1 + 2R_3 \\ R_2 - 3R_3 \end{array}} \left[ \begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]. \end{aligned}$ |
| basis: inferred / ink                                                                                                                                                                                                                                                                                               |      |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| johnston-linear-matrix-algebra-EQ1276 (was)                                                                                                                                                                                                                                                                         | 427  | ■0.022 | $\left. 3x^2 - 5x - 2 \right) \frac{x^2 + 3x + 1}{3x^4 + 4x^3 - 14x^2 - 11x - 2} = \frac{3x^4 - 5x^3 - 2x^2}{9x^3 - 12x^2 - 11x - 2} + \frac{9x^3 - 15x^2 - 6x}{3x^2 - 5x - 2} + \frac{3x^2 - 5x - 2}{0}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\left. 3x^2 - 5x - 2 \right) \frac{x^2 + 3x + 1}{3x^4 + 4x^3 - 14x^2 - 11x - 2} = \frac{3x^4 - 5x^3 - 2x^2}{9x^3 - 12x^2 - 11x - 2} + \frac{9x^3 - 15x^2 - 6x}{3x^2 - 5x - 2} + \frac{3x^2 - 5x - 2}{0}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\begin{aligned} & \frac{x^2 + 3x + 1}{3x^2 - 5x - 2} \overline{) 3x^4 + 4x^3 - 14x^2 - 11x - 2} \\ & \underline{3x^4 - 5x^3 - 2x^2} \\ & 9x^3 - 12x^2 - 11x - 2 \\ & \underline{9x^3 - 15x^2 - 6x} \\ & 3x^2 - 5x - 2 \\ & \underline{3x^2 - 5x - 2} \\ & 0 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <div>Conf. MathPix confidence: <span>■</span> ≥0.9 <span>■</span> 0.5-0.9 <span>■</span> &lt;0.5 — no value</div> <div>Residual render vs scan (inkdrill): <span>●</span> C component <span>●</span> W weak <span>●</span> S stable <span>●</span> N noise <span>●</span> K clean <span>●</span> not measured</div> |      |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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| johnston-linear-matrix-attached<br>E01276 (now)                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 427  | —     | \begin{array}{r} x^2+3x+1\phantom{xxxxxxxx}\backslash\\ \backslash 3x^2-5x-2\overline{)3x^4+4x^3-14x^2-11x-2}\backslash\backslash\\ \underline{3x^4-5x^3-2x^2\phantom{xxxxxxxx}}\backslash\backslash\\ 9x^3-12x^2-11x-2\backslash\backslash \underline{9x^3-15x^2-6x\phantom{xxx}}\backslash\backslash\\ \backslash 3x^2-5x-2\backslash\backslash \underline{3x^2-5x-2}\backslash\backslash 0 \end{array} | $ \begin{array}{r} x^2 + 3x + 1 \\ 3x^2 - 5x - 2 \overline{) 3x^4 + 4x^3 - 14x^2 - 11x - 2} \\ \underline{3x^4 - 5x^3 - 2x^2} \phantom{- 11x - 2} \\ 9x^3 - 12x^2 - 11x - 2 \\ \underline{9x^3 - 15x^2 - 6x} \phantom{- 2} \\ 3x^2 - 5x - 2 \\ \underline{3x^2 - 5x - 2} \\ 0 \end{array} $ |  |
| basis: inferred / ink                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |       |                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                             |                                                                                     |
| <b>Conf.</b> MathPix confidence: <span style="color: green;">■</span> $\geq 0.9$ <span style="color: orange;">■</span> 0.5-0.9 <span style="color: red;">■</span> $< 0.5$ — no value<br><b>Residual</b> render vs scan (inkdrill): <span style="color: red;">●</span> C component <span style="color: orange;">●</span> W weak <span style="color: blue;">●</span> S stable <span style="color: grey;">●</span> N noise <span style="color: green;">●</span> K clean <span style="color: grey;">●</span> not measured |      |       |                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                             |                                                                                     |

Unresolved

| Identifier                                                                                                                                                                                                                                                                                                                                                      | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Image |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| johnston-linear-matrix-algebra-EQ0114 [conf 0.025]                                                                                                                                                                                                                                                                                                              | 63   | 0.025 | <pre>\begin{figure} \[ R^{\theta}\left(\mathbf{e}_{2}\right)=(-\sin (\theta), \cos (\theta)) \overbrace{\cos (\theta)}^{-\sin (\theta)} \overbrace{\cos (\theta)}^{\sin (\theta)} \mathbf{e}_{1}^{\mathbf{e}_{2}} \times \text { - } \underbrace{\cos (\theta)}_{R^{\theta}\left(\mathbf{e}_{1}\right)} \left(\mathbf{e}_{1}\right)=\left(\cos (\theta), \sin (\theta)\right) \]</pre> <p>Figure 1.19: <math>(R^{\theta})</math> rotates the standard basis vectors <math>(\mathbf{e}_1)</math> and <math>(\mathbf{e}_2)</math> to <math>(R^{\theta}(\mathbf{e}_1))</math> and <math>(R^{\theta}(\mathbf{e}_2))</math> to <math>(\cos (\theta), \sin (\theta))</math> and <math>(-\sin (\theta), \cos (\theta))</math>, respectively.</p>                                                                                                                                                                                                                                            | (not rendered)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |
| johnston-linear-matrix-algebra-EQ0947 [conf 0.100]                                                                                                                                                                                                                                                                                                              | 336  | 0.100 | <pre>\begin{array}{ll} \mathbf{p}_1 \mathbf{q}_1^{*}=\frac{1}{3}\left[\begin{array}{l} 3 \\ 1 \\ 0 \\ 0 \end{array}\right]\left[\begin{array}{lll} 1 &amp; 0 &amp; -1 \end{array}\right]=\frac{1}{3}\left[\begin{array}{lll} 3 &amp; 0 &amp; -3 \\ 1 &amp; 0 &amp; -1 \\ 0 &amp; 0 &amp; 0 \\ 0 &amp; 0 &amp; 0 \end{array}\right] \text { pare } \\ \mathbf{p}_2 \mathbf{q}_2^{*}=\frac{1}{3}\left[\begin{array}{l} 1 \\ 0 \\ 0 \\ 1 \end{array}\right]\left[\begin{array}{lll} -1 &amp; 3 &amp; 1 \end{array}\right]=\frac{1}{3}\left[\begin{array}{lll} 0 &amp; 0 &amp; 0 \\ -1 &amp; 3 &amp; 1 \\ 0 &amp; 0 &amp; 0 \\ 0 &amp; 0 &amp; 0 \end{array}\right], \text { and } \\ \mathbf{p}_3 \mathbf{q}_3^{*}=\frac{1}{3}\left[\begin{array}{l} 1 \\ 0 \\ 1 \end{array}\right]\left[\begin{array}{lll} 0 &amp; 0 &amp; 3 \end{array}\right]=\frac{1}{3}\left[\begin{array}{lll} 0 &amp; 0 &amp; 3 \\ 0 &amp; 0 &amp; 0 \\ 0 &amp; 0 &amp; 3 \end{array}\right] . \end{array}</pre> | $\mathbf{p}_1 \mathbf{q}_1^{*}=\frac{1}{3}\left[\begin{array}{l} 3 \\ 1 \\ 0 \end{array}\right]\left[\begin{array}{lll} 1 & 0 & -1 \end{array}\right]=\frac{1}{3}\left[\begin{array}{lll} 3 & 0 & -3 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{array}\right]$ $\mathbf{p}_2 \mathbf{q}_2^{*}=\frac{1}{3}\left[\begin{array}{l} 0 \\ 1 \\ 0 \end{array}\right]\left[\begin{array}{lll} -1 & 3 & 1 \end{array}\right]=\frac{1}{3}\left[\begin{array}{lll} 0 & 0 & 0 \\ -1 & 3 & 1 \\ 0 & 0 & 0 \end{array}\right], \text { and }$ $\mathbf{p}_3 \mathbf{q}_3^{*}=\frac{1}{3}\left[\begin{array}{l} 1 \\ 0 \\ 1 \end{array}\right]\left[\begin{array}{lll} 0 & 0 & 3 \end{array}\right]=\frac{1}{3}\left[\begin{array}{lll} 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{array}\right] .$ |       |
| <div>Conf. MathPix confidence: <span>■</span> <math>\geq 0.9</math> <span>■</span> <math>0.5-0.9</math> <span>■</span> <math>&lt; 0.5</math> — no value</div> <div>Residual render vs scan (inkdrill): <span>●</span> C component <span>●</span> W weak <span>●</span> S stable <span>●</span> N noise <span>●</span> K clean <span>●</span> not measured</div> |      |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |

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| johnston-linear-matrix-algebra-E09439                                                                                                                                                                                                                                                                                                                                                  | 170  | <div> <div>0.131</div> <div></div> </div> | <pre> \begin{figure} \[ \begin{aligned} {[A \mid I] } &amp;= \left[ \begin{array}{cccccc cccc} * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; 1 &amp; 0 &amp; 0 &amp; 0 \\ * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; 0 &amp; 1 &amp; 0 &amp; 0 \\ * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; 0 &amp; 0 &amp; 1 &amp; 0 \\ * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; 0 &amp; 0 &amp; 0 &amp; 1 \end{array} \right] \\ \xrightarrow{\text{row-reduce}} &amp; \left[ \begin{array}{cccccc cccc} \star &amp; * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; * \\ 0 &amp; 0 &amp; \star &amp; * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; * &amp; * \\ 0 &amp; 0 &amp; 0 &amp; 0 &amp; 0 &amp; \star &amp; * &amp; * &amp; * &amp; * &amp; * &amp; * \\ 0 &amp; 0 &amp; 0 &amp; 0 &amp; 0 &amp; 0 &amp; 0 &amp; 0 &amp; * &amp; * &amp; * &amp; * \end{array} \right] = [R \mid E] \\ &amp; \leftarrow \text{null}(A^T) \end{aligned} \]</pre> <p>After we row-reduce <math>[[A \mid I]]</math> to a row echelon form <math>[[R \mid E]]</math>, we can immediately read off bases of <math>(\text{range}(A^T))</math> and <math>(\text{null}(A^T))</math> from the rows of that matrix.</p> | (not rendered)                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                     |
| johnston-linear-matrix-algebra-E09294                                                                                                                                                                                                                                                                                                                                                  | 98   | <div> <div>0.137</div> <div></div> </div> | <pre> \begin{aligned} \text{ (equation 3) } &amp; 2x + 4y + 5z = 12 \\ -2(\text{equation 1}) &amp; -2x - 6y + 4z = -10 \\ \hline \text{ (new equation 3) } &amp; -2y + 9z = 2 \end{aligned}</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <div> <div>(equation 3)</div> <div><math>2x + 4y + 5z = 12</math></div> </div> <div> <div><math>-2(\text{equation 1})</math></div> <div><math>-2x - 6y + 4z = -10</math></div> </div> <div> <div>(new equation 3)</div> <div><math>-2y + 9z = 2</math></div> </div>                                 | <div> <div>(equation 3)</div> <div><math>2x + 4y + 5z = 12</math></div> </div> <div> <div><math>-2(\text{equation 1})</math></div> <div><math>-2x - 6y + 4z = -10</math></div> </div> <div> <div>(new equation 3)</div> <div><math>-2y + 9z = 2</math></div> </div>                                 |
| johnston-linear-matrix-algebra-E01955                                                                                                                                                                                                                                                                                                                                                  | 361  | <div> <div>0.311</div> <div></div> </div> | <pre> \begin{array}{lll} \left( \begin{array}{lll} 1 &amp; 2 &amp; 3 \\ 1 &amp; 3 &amp; 2 \end{array} \right) &amp; \left( \begin{array}{lll} 2 &amp; 1 &amp; 3 \\ 2 &amp; 3 &amp; 1 \end{array} \right) &amp; \left( \begin{array}{lll} 3 &amp; 1 &amp; 2 \\ 3 &amp; 2 &amp; 1 \end{array} \right) \end{array}</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <div> <div><math>\begin{pmatrix} 1 &amp; 2 &amp; 3 \\ 1 &amp; 3 &amp; 2 \end{pmatrix}</math></div> <div><math>\begin{pmatrix} 2 &amp; 1 &amp; 3 \\ 2 &amp; 3 &amp; 1 \end{pmatrix}</math></div> <div><math>\begin{pmatrix} 3 &amp; 1 &amp; 2 \\ 3 &amp; 2 &amp; 1 \end{pmatrix}</math></div> </div> | <div> <div><math>\begin{pmatrix} 1 &amp; 2 &amp; 3 \\ 1 &amp; 3 &amp; 2 \end{pmatrix}</math></div> <div><math>\begin{pmatrix} 2 &amp; 1 &amp; 3 \\ 2 &amp; 3 &amp; 1 \end{pmatrix}</math></div> <div><math>\begin{pmatrix} 3 &amp; 1 &amp; 2 \\ 3 &amp; 2 &amp; 1 \end{pmatrix}</math></div> </div> |
| <div> <div>Conf.</div> <div>MathPix confidence:</div> <div> <div>■ ≥0.9</div> <div>■ 0.5-0.9</div> <div>■ &lt;0.5</div> <div>— no value</div> </div> </div> <div> <div>Residual</div> <div>render vs scan (inkdrill):</div> <div> <div>● C component</div> <div>● W weak</div> <div>● S stable</div> <div>● N noise</div> <div>● K clean</div> <div>● not measured</div> </div> </div> |      |                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                     |

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| johnston-linear-matrix-algebra-<br>EQ0575                                                                                                                                              | 224  | ■ 0.324<br>● C +93 | <pre>\begin{figure} \[ \left.A=\left[\begin{array}{llllll} * &amp; * &amp; * &amp; * &amp; * &amp; * \\ * &amp; * &amp; * &amp; * &amp; * &amp; * \\ * &amp; * &amp; * &amp; * &amp; * &amp; * \\ * &amp; * &amp; * &amp; * &amp; * &amp; * \\ * &amp; * &amp; * &amp; * &amp; * &amp; * \\ * &amp; * &amp; * &amp; * &amp; * &amp; * \end{array}\right]=\left[\begin{array}{lll} * &amp; * &amp; * \\ * &amp; * &amp; * \\ * &amp; * &amp; * \\ * &amp; * &amp; * \\ * &amp; * &amp; * \\ * &amp; * &amp; * \end{array}\right] \left[\begin{array}{llllll} * &amp; * &amp; * &amp; * &amp; * &amp; * \\ * &amp; * &amp; * &amp; * &amp; * &amp; * \\ * &amp; * &amp; * &amp; * &amp; * &amp; * \\ * &amp; * &amp; * &amp; * &amp; * &amp; * \\ * &amp; * &amp; * &amp; * &amp; * &amp; * \\ * &amp; * &amp; * &amp; * &amp; * &amp; * \end{array}\right] \]</pre> <p>(not rendered)</p> |          |       |
| johnston-linear-matrix-algebra-<br>EQ0218                                                                                                                                              | 102  | ■ 0.407<br>● U -9  | <pre>\left[\begin{array}{ccccc} 0 &amp; 2 &amp; -2 &amp; 6 &amp; -3 \\ 0 &amp; -2 &amp; 1 &amp; -5 &amp; 2 \\ 0 &amp; 0 &amp; -1 &amp; 1 &amp; 0 \\ 0 &amp; -4 &amp; 2 &amp; -10 &amp; 5 \end{array}\right] \xrightarrow[R_4+2R_1]{R_2+R_1} \left[\begin{array}{ccccc} 0 &amp; 2 &amp; -2 &amp; 6 &amp; -3 \\ 0 &amp; 0 &amp; -1 &amp; 1 &amp; 0 \\ 0 &amp; 0 &amp; -1 &amp; 1 &amp; 0 \\ -2 &amp; 2 &amp; -1 &amp; 0 &amp; 0 \end{array}\right]</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |       |
| <b>Conf.</b> MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value<br><b>Residual</b> render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured |      |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |       |

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| johnston-linear-matrix-algebra-EQ0884                                                                                                                                                  | 318  | ■ 0.451<br>● C +38  | $\begin{aligned} \operatorname{rank}(A) &= \operatorname{rank}(B) = 3, \quad \operatorname{tr}(A) = \operatorname{tr}(B) = 0, \\ \det(A) &= \det(B) = 2, \quad \text{and} \quad p_A(\lambda) = p_B(\lambda) = -\lambda^3 + 3\lambda + 2. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                  | $\operatorname{rank}(A) = \operatorname{rank}(B) = 3, \quad \operatorname{tr}(A) = \operatorname{tr}(B) = 0, \dots$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\operatorname{rank}(A) = \operatorname{rank}(B) = 3, \quad \operatorname{tr}(A) = \operatorname{tr}(B) = 0, \dots$<br>$\det(A) = \det(B) = 2, \quad \text{and} \quad p_A(\lambda) = p_B(\lambda) = -\lambda^3 + 3\lambda + 2.$                                                                                                                                                                                                                                                                                                                                                                                                                         |
| johnston-linear-matrix-algebra-EQ0793                                                                                                                                                  | 293  | ■ 0.462<br>● C +130 | $\left[ \begin{array}{cccc} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ 0 & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & a_{n,2} & \cdots & a_{n,n} \end{array} \right] \quad \text{has row echelon form} \quad \left[ \begin{array}{c ccc} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ \hline 0 & & & \\ \vdots & & & \\ 0 & & & R \end{array} \right],$                                                                                                                                                                                                                                                                              | $\left[ \begin{array}{c ccc} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ \hline 0 & & & \\ \vdots & & & \\ 0 & & & R \end{array} \right],$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\left[ \begin{array}{c ccc} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ \hline 0 & & & \\ \vdots & & & \\ 0 & & & R \end{array} \right],$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| johnston-linear-matrix-algebra-EQ0040                                                                                                                                                  | 39   | ■ 0.479<br>● C +817 | $\begin{aligned} A B &= \left[ \begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right] \left[ \begin{array}{lll} 5 & 6 & 7 \\ 8 & 9 & 10 \end{array} \right] \\ &= \left[ \begin{array}{lll} (1,2) \cdot (5,8) & (1,2) \cdot (6,9) & (1,2) \cdot (7,10) \\ (3,4) \cdot (5,8) & (3,4) \cdot (6,9) & (3,4) \cdot (7,10) \end{array} \right] \\ &= \left[ \begin{array}{lll} 1 \times 5 + 2 \times 8 & 1 \times 6 + 2 \times 9 & 1 \times 7 + 2 \times 10 \\ 3 \times 5 + 4 \times 8 & 3 \times 6 + 4 \times 9 & 3 \times 7 + 4 \times 10 \end{array} \right] \\ &= \left[ \begin{array}{lll} 21 & 24 & 27 \\ 47 & 54 & 61 \end{array} \right]. \end{aligned}$ | $\begin{aligned} AB &= \left[ \begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right] \left[ \begin{array}{lll} 5 & 6 & 7 \\ 8 & 9 & 10 \end{array} \right] \\ &= \left[ \begin{array}{lll} (1,2) \cdot (5,8) & (1,2) \cdot (6,9) & (1,2) \cdot (7,10) \\ (3,4) \cdot (5,8) & (3,4) \cdot (6,9) & (3,4) \cdot (7,10) \end{array} \right] \\ &= \left[ \begin{array}{lll} 1 \times 5 + 2 \times 8 & 1 \times 6 + 2 \times 9 & 1 \times 7 + 2 \times 10 \\ 3 \times 5 + 4 \times 8 & 3 \times 6 + 4 \times 9 & 3 \times 7 + 4 \times 10 \end{array} \right] \\ &= \left[ \begin{array}{lll} 21 & 24 & 27 \\ 47 & 54 & 61 \end{array} \right]. \end{aligned}$ | $\begin{aligned} AB &= \left[ \begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right] \left[ \begin{array}{lll} 5 & 6 & 7 \\ 8 & 9 & 10 \end{array} \right] \\ &= \left[ \begin{array}{lll} (1,2) \cdot (5,8) & (1,2) \cdot (6,9) & (1,2) \cdot (7,10) \\ (3,4) \cdot (5,8) & (3,4) \cdot (6,9) & (3,4) \cdot (7,10) \end{array} \right] \\ &= \left[ \begin{array}{lll} 1 \times 5 + 2 \times 8 & 1 \times 6 + 2 \times 9 & 1 \times 7 + 2 \times 10 \\ 3 \times 5 + 4 \times 8 & 3 \times 6 + 4 \times 9 & 3 \times 7 + 4 \times 10 \end{array} \right] \\ &= \left[ \begin{array}{lll} 21 & 24 & 27 \\ 47 & 54 & 61 \end{array} \right]. \end{aligned}$ |
| johnston-linear-matrix-algebra-EQ0073                                                                                                                                                  | 48   | ■ 0.493<br>● C +396 | $\begin{aligned} A B &= \left[ \begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right] \left[ \begin{array}{ccc} -1 & 1 & 1 \\ 0 & 1 & 0 \end{array} \right] \\ &= \left[ \begin{array}{ccc} (1,2) \cdot (-1,0) & (1,2) \cdot (1,1) & (1,2) \cdot (1,0) \\ (3,4) \cdot (-1,0) & (3,4) \cdot (1,1) & (3,4) \cdot (1,0) \end{array} \right] \\ &= \left[ \begin{array}{ccc} -1 & 3 & 1 \\ -3 & 7 & 3 \end{array} \right]. \end{aligned}$                                                                                                                                                                                                                      | $\begin{aligned} AB &= \left[ \begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right] \left[ \begin{array}{ccc} -1 & 1 & 1 \\ 0 & 1 & 0 \end{array} \right] \\ &= \left[ \begin{array}{ccc} (1,2) \cdot (-1,0) & (1,2) \cdot (1,1) & (1,2) \cdot (1,0) \\ (3,4) \cdot (-1,0) & (3,4) \cdot (1,1) & (3,4) \cdot (1,0) \end{array} \right] \\ &= \left[ \begin{array}{ccc} -1 & 3 & 1 \\ -3 & 7 & 3 \end{array} \right]. \end{aligned}$                                                                                                                                                                                                                      | $\begin{aligned} AB &= \left[ \begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right] \left[ \begin{array}{ccc} -1 & 1 & 1 \\ 0 & 1 & 0 \end{array} \right] \\ &= \left[ \begin{array}{ccc} (1,2) \cdot (-1,0) & (1,2) \cdot (1,1) & (1,2) \cdot (1,0) \\ (3,4) \cdot (-1,0) & (3,4) \cdot (1,1) & (3,4) \cdot (1,0) \end{array} \right] \\ &= \left[ \begin{array}{ccc} -1 & 3 & 1 \\ -3 & 7 & 3 \end{array} \right]. \end{aligned}$                                                                                                                                                                                                                      |
| <b>Conf.</b> MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value<br><b>Residual</b> render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |